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Machine Learning-based Energy-efficient Workload Management for Data Centers

Matthew Smith*, Luke Zhao†, Jonathan Cordova*, Xunfei Jiang*, and Mahdi Ebrahimi*
Department of Computer Science, California State University, Northridge *
Department of Computer Science, Brown University †
matthew.smith.227@my.csun.edu, luke_zhao@brown.edu, jonathan.cordova.394@my.csun.edu,
{xunfei.jiang, mahdi.ebrahimi}@csun.edu

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***Abstract*—Cooling costs count for a significant part of the total energy consumption in data centers, and previous researchers mainly focused on investigating thermal-ware workload distribution strategies for CPU-intensive workloads. This paper introduces a novel machine learning-based approach that aims at reducing energy consumption through thermal-aware workload distribution to build energy-efficient data centers for GPU-intensive workload. To achieve this goal, the study employs the GPUCloudSim Plus simulator, which effectively models the distribution of GPU-intensive applications under diverse workloads and utilizations. The integration of advanced machine learning models allows for accurate temperature predictions and comprehensive evaluation of the proposed algorithm's performance. We evaluated our ThermalAwareGpu workload scheduling algorithm, and it saved up to 12.82% of computing cost compared to the baseline algorithms. Our future work will explore the estimation of data center cooling energy and conduct in-depth comparisons of different workload balancing algorithms on more intensive experiments.**

***Index Terms*—machine learning, energy-efficient, workload management, data centers**

I. Introduction

Datacenters consume about 1% of the world's energy, 40% of which is estimated to be from the cost of cooling server hardware [1]. Thermal aware workload balancing strategies have been shown to reduce CPU temperature within datacenters [2]. Most research on energy-efficient load balancing focuses on virtual machine migration. The process of migrating a virtual machine consumes resources when transferring data from one host to another.

We proposed a machine learning-based approach to optimize energy-efficient workload management in data centers. Leveraging the GpuCloudSim Plus simulator, we effectively modeled the distribution of GPU-intensive applications under diverse workloads and utilizations. By strategically distributing GPU-intensive tasks across servers, we aim to maintain high system performance and efficient resource utilization, while minimizing both computing and cooling costs. We integrated advanced machine learning models that enable precise temperature predictions. We proposed a GPU thermal-aware approach for GPU-intensive workload distribution and evaluated the performance and energy consumption of our algorithm using three different types of GPU-intensive workloads. Our ThermalAwareGpu algorithm saved up to 12.82% of computing cost, comparing with the other baseline algorithms.

The rest of this paper is organized as following: Section II discusses related work; Section III presents the design of the project; Section IV presents the experimental results; and Section V concludes the project and discusses future work.

II. Related Works

A. Data Center Simulators

Cloud simulation is widely used to test and evaluate new models and workload management algorithms. GreenCloud is a data center simulator that was built for energy-aware cloud computing [3]. It could simulate workload distribution and capture energy consumption data for components (servers, switches, and links) and packet-level communication patterns in realistic setups. It also provides a friendly user interface with visualized simulation results. However, it was discontinued and is not publicly accessible. CloudSim [4] is an extensible simulator toolkit that supporting modeling and simulation of cloud computing systems and application provisioning. It supports system and behavior modeling of data centers, virtual machines, and resource provisioning policies and simulations of cloud computing environments, including single and inter-networked clouds. There have been many simulators that are derived from CloudSim with advanced features. CloudAnalyst simulates large-scale Cloud applications to investigate large-scale Cloud applications in geographic distribution of both computing servers and user workloads [5]. CloudSim Plus [6] is a highly configurable and extensible cloud computing simulator that aims to improve maintainability, reusability and extensibility. With the increasing usage of GPUs in cloud data centers, there was an urgent need for a simulation environment that is capable to study the effectiveness of GPU virtualization techniques. GpuCloudSim was built [7] to model and simulate GPU-enabled VMs in cloud data centers. It can simulate interference among co-running applications, the overhead of virtualization, and power consumption of GPUs to evaluate GPU provisioning and scheduling policies in cloud data centers. An agent-based cloud simulation model was recently introduced for resource management, with main cloud elements are modeled as autonomous agents whose properties are encapsulated [8]. It focuses on the effect of service placement strategies, service migration, and server consolidation on the performance of clouds, in terms of energy

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consumption, resource utilization, and violation of service-level agreements.

B. Thermal-aware Workload Scheduling for Data Centers

There have been increasing research study the reduction of cooling cost in data centers. Moore et al. developed real-world scheduling algorithms, zone-based discretization (ZBD) and minimize-heat-recirculation (MinHR) [9]. Their research demonstrated up to a factor of two reduction in annual data center cooling costs over location-agnostic workload distribution. Guo et al. raised a temperature-aware Virtual Data Center (VDC) embedding scheme to reduce hot spots by minimizing the maximum temperature of hot air emitted from each rack, which in consequence reduces power consumption of IT equipments [10]. A Mixed Integer Linear Programming (MILP) model and a heuristic algorithm were implemented for the proposed VDC embedding scheme. Ilager et al. introduced a dynamic scheduling algorithm to minimize the peak temperature of servers [2]. Compared with their baseline algorithm, they reduced the peak temperature by 6.5 °C and consumed 34.5% less energy. Li et al. proposed a new holistic model to characterize the impacts of workloads, computing and cooling costs, heat recirculation, and node failure on the energy efficiency of cloud data centers, and implemented a thermal-aware workload distribution algorithm to improve the energy efficiency of cloud data centers which lowered the cooling cost by 15.4% to 50% [11].

III. Design

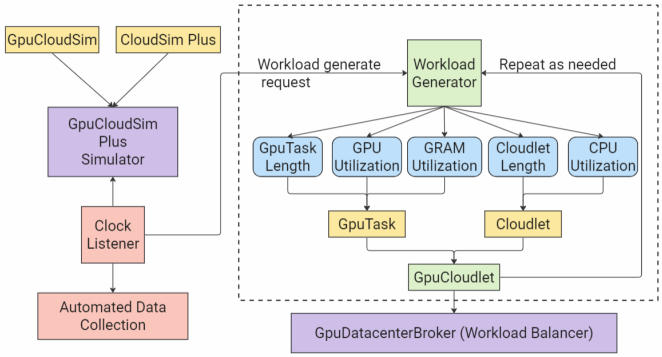


Fig. 1: Framework for the Project Simulator

As shown in Fig. 1, GpuCloudSim and CloudSim Plus are integrated to form the GpuCloudSim Plus simulator. An event listener is registered in the simulator that will be triggered when the simulated time changes. Every second of the simulation, the listener will invoke both the workload generator and automated data collection modules. Once a request is sent by the clock listener, the workload generator will create tasks, GpuCloudlets, and submit them to the simulator. A GpuCloudlet is composed of a Cloudlet object and a GpuTask object, which represent CPU-intensive and GPU-intensive tasks, respectively. Uniform random distributions are used to generate GpuTask and Cloudlet lengths, GPU utilization, GPU RAM (GRAM) utilization, and CPU utilization. These metrics are used by the simulator to simulate the execution of each GpuCloudlet. The workload balancer (GpuDatacenterBroker) receives incoming tasks from the workload generator and allocates them to available virtual machines. The automated data collection module is designed to collect metric data that are associated with the simulation. The metrics are processed by the module and a formatted output file will be generated for simulation analysis.

A. Workload Generator

The workload supports generation for three common workload patterns in data centers: random, intermittent, and periodic [12]. For random workloads, there will be a random number of Cloudlets been generated every second. In intermittent workloads, small numbers of tasks will be followed by a burst of incoming tasks. For periodic workloads, tasks are scheduled at specified intervals – this type of data center utilization closely resembles a sine wave.

B. Automated Data Collection

The automated data collection module acts as clock event listener for the GpuCloudSim Plus simulator. This listener waits until a simulated second has elapsed, then it retrieves the following metric data for each host server: CPU and GPU power, utilization for CPU, GPU, and GRAM, and GPU temperature. Average values of these metrics for each server are calculated and saved.

When a user runs a set of experiments to compare the performance and energy efficiency of using different algorithms to schedule various workload types, the metric data will be written to a series of CSV files after all the simulations are completed. These CSV files are automatically merged into a workload-specific Excel document to show the summarized comparison of all scheduling algorithms. An overall comparison sheet is generated to display metric data, for an entire experiment, in tabular formats.

C. Thermal-Aware Workload Schedulers

Besides the built-in workload scheduler algorithms, the two additional algorithms that are implemented for this project are as following:

1) ThermalAwareBaseline: A thermal-aware algorithm designed for CPU virtual machine migration [2] was modified to distribute tasks for GPU-intensive applications. This algorithm uses both utilization and temperature thresholds to prevent hosts from becoming overutilized and too warm. The thresholds in the adapted algorithm are set to 50°C for GPU temperature and 75% GPU utilization.

2) ThermalAwareGPU: The algorithm [12] was designed to quickly find available virtual machines in servers that have the lowest GPU temperatures. It also aims to reduce the overhead by retrieving servers' temperatures at a specified time interval, rather than requiring real-time temperature readings. The algorithm was previously evaluated at a 10 second interval.

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IV. Results

Using our modified GpuCloudSim Plus simulator[1], two sets of experiments were designed to evaluate the performance of the ThermalAwareGPU algorithm by comparing with several baseline algorithms. The configuration of the computer server for running the simulations is shown in Table I.

TABLE I: Computer Server Configuration

CPU	2 * Intel(R) Xeon(R) CPU E5-2690 v2 @ 3.00GHz (10 Cores)
Memory	98,304 MB
Disk	2 * 480 GB
GPU	NVIDIA Tesla P4 8GB GDDR5
OS	Ubuntu 22.04.2 LTS

We simulated a data center with four hosts. Each host was configured with 20 CPU cores and 500 GPU cores. Five virtual machines were created on each host, with each virtual machine having 4 CPU cores (50,000 MIPS capacity) and 100 GPU cores (50,000 MIPS capacity). We scaled the MIPS capacity of a single CPU core from the default value (1,000 MIPS) upwards to 12,500 MIPS to avoid issues with the generated Cloudlets not finishing properly in the GpuCloudSim Plus simulator. We ran each simulation for one hour, and expanded on our previous work to compare the performance of different time intervals for sorting the servers based on GPU temperatures in the ThermalAwareGPU algorithm.

A. Experiment Set 1

Uniform random distributions were used to set GPU utilization between 25%–95% and GRAM utilization between 5%–95% for each task in the workload generator. These bounds were chosen to match the span of the benchmark training data used for our machine learning models. As shown in Table II, the ThermalAwareBaseline (TABaseline) algorithm was consistently one of the highest average GPU temperature. The performance of ThermalAwareGPU (TAGpu) was not generalizable across different workloads.

TABLE II: Set 1 Average GPU Temperature (°C)

GPU Temp. Sort Interval	Algorithm Name	Workload Type		
		Random	*Periodic*	*Intermittent*
1 Second	BestFit	31.33	31.23	31.48
	FirstFit	31.34	31.21	31.52
	RoundRobin	31.33	31.21	31.52
	TABaseline	31.49	31.49	31.48
	TAGpu	31.38	31.31	31.37
5 Seconds	BestFit	31.32	31.23	31.48
	FirstFit	31.32	31.23	31.53
	RoundRobin	31.22	31.23	31.53
	TABaseline	31.50	31.49	31.48
	TAGpu	31.34	31.26	31.29
10 Seconds	BestFit	31.27	31.24	31.49
	FirstFit	31.28	31.24	31.53
	RoundRobin	31.28	31.24	31.53
	TABaseline	31.46	31.23	31.48
	TAGpu	31.31	31.26	31.29

These irregularities likely stem from incorrect temperature predictions caused by limited model training datasets, randomness in the workload task generation, and the assumption that all tasks will significantly heat up the GPU [12].

[1]https://github.com/CSUN-REU/GPUCloudSimPlus

TABLE III: Set 1 Computing Energy Cost Comparison (KJs)

GPU Temp. Sort Interval	Algorithm Name	Workload Type		
		Random	*Periodic*	*Intermittent*
1 Second	BestFit	999.64	994.20	993.68
	FirstFit	1017.69	1006.51	1010.04
	RoundRobin	1018.00	1006.49	1010.23
	TABaseline	903.17	905.01	899.18
	TAGpu	918.11	922.37	912.79
5 Seconds	BestFit	1000.60	994.41	999.12
	FirstFit	1017.81	1008.61	1006.98
	RoundRobin	1017.81	1009.52	1007.52
	TABaseline	902.50	907.06	897.29
	TAGpu	909.78	911.50	897.52
10 Seconds	BestFit	1002.23	994.43	989.30
	FirstFit	1020.05	1007.79	1004.95
	RoundRobin	1020.05	1007.64	1005.18
	TABaseline	907.12	907.51	899.72
	TAGpu	910.95	908.19	893.51

Table III shows the computational energy cost for experiment set 1. For all workloads, there were improved energy savings when increasing the temperature sort time interval. With the intervals of 1 and 5 seconds, the ThermalAwareGPU consumed 1.70% and 0.42% more energy on average than the ThermalAwareBaseline algorithm. For the 10-second time interval, the ThermalAwareGPU saved 0.07% compared to the ThermalAwareBaseline. Energy savings for ThermalAwareGPU ranged from 7.84% to 9.19% when compared to the next best performing algorithm.

B. Experiment Set 2

GRAM utilization was set between 50%–100% and GPU utilization was increased to 75%–100% to simulate more GPU-intense tasks [12]. The ThermalAwareGPU algorithm outperformed all the baseline algorithms across all experiments in this set. Table IV shows the average GPU temperature using the two thermal-aware scheduling algorithms and the next best performance algorithm, FirstFit.

TABLE IV: Set 2 Average GPU Temperature (°C)

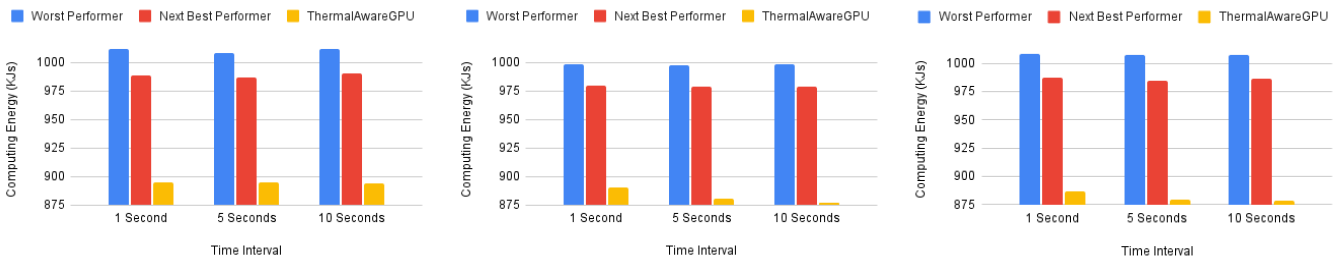
GPU Temp. Sort Interval	Algorithm Name	Workload Type		
		Random	*Periodic*	*Intermittent*
1 Second	FirstFit	31.93	31.85	31.94
	TABaseline	31.71	31.70	31.68
	TAGpu	31.68	31.64	31.63
5 Seconds	FirstFit	31.91	31.84	31.92
	TABaseline	31.71	31.70	31.66
	TAGpu	31.65	31.57	31.55
10 Seconds	FirstFit	31.91	31.83	31.93
	TABaseline	31.71	31.71	31.67
	TAGpu	31.64	31.55	31.55

Compared to the ThermalAwareBaseline algorithm, our ThermalAwareGPU scheduling algorithm lowered the GPU temperature by 0.15%, 0.32%, 0.37%, for 1, 5, and 10 second intervals respectively. Compared to FirstFit, the next best performing algorithm, ThermalAwareGPU lowered the GPU temperature by 0.80%, 0.94%, and 0.97% for the three temperature sorting time intervals. A larger temperature reduction occurred when changing the time interval from a 1 second to 5 seconds, where changing from 5 to 10 seconds showed less improvement. Temperature reduction using ThermalAwareGPU

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(a) Random workload (b) Periodic workload (c) Intermittent workload

Fig. 2: Computing Energy Cost under Various Time Internals for Different Workloads

had the least improvement for random workload, comparing with other two types of workloads. This could be explained by the random workload's higher utilization of virtual machines, leaving less room for tasks to be allocated to cooler hosts.

ThermalAwareGPU was mostly the best algorithm, comparing with the other baseline algorithms. Though it consumed 0.09% more energy than ThermalAwareBaseline for 1-second sorting time interval, it saved 0.57% and 0.78% more energy than ThermalAwareBaseline in 5 and 10 seconds intervals. It reduced the computing cost by up to 1.28% comparing with the ThermalAwareBaseline algorithm. Fig. 2 illustrates the computing energy consumption of ThermalAwareGPU, the next best algorithm, and the worst performing algorithm. Our ThermalAwareGPU algorithm showed energy savings of 9.99% on average and up to 12.82% when compared to load balancing algorithms that do not consider temperature.

C. Analysis

ThermalAwareGPU outperformed ThermalAwareBaseline with maximum power savings of 1.35% and temperature reduction of 0.50% in some circumstances. For both experiment sets, increasing ThermalAwareGPU's time interval from 1 up to 10 seconds led to larger reductions in both computing energy and average GPU temperature. Scheduling the higher GPU-intensive tasks using ThermalAwareGPU algorithm saved more computing energy. With maximal reduction in temperature and computing energy consumption for both experiment sets, 10 second temperature sorting interval is an optimal configuration for ThermalAwareGPU.

V. Conclusion and Future Work

We conducted experiments to find an optimal sorting time interval for our ThermalAwareGPU algorithm, and evaluated its efficiency in energy saving and temperature reduction with four baseline algorithms under three different workload types. Our algorithm saved up to 12.82% energy compared to non-thermal-aware scheduling algorithms.

Future work will be conducted to improve the prediction of the GPU temperature and power models. We will evaluate the performance of the ThermalAwareGPU algorithm on temperature sorting time intervals above 10 seconds to see if energy can be further reduced. We will also simulate larger data centers with more hosts and virtual machines and analyze our algorithm's performance on real-world workload traces. The energy saved for cooling GPU and server will also be estimated and considered in the algorithm evaluation.

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